

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

Perfomax

May 2026–present

AI Advisor

- **AI Strategy & Platform Architecture :**

- Advised on AI/ML strategy, product implementation, and technical design for the platform.

National University of Singapore

2022–present

Postdoc under Kostya Novoselov

- **Leadership :**

- Speaker search and selection for the 500-person AI4X 2025 conference.
- Main organizer of the ICLR 2025 workshop on multiscale machine learning.

- **Transformer for generating symmetric crystals :**

- Created a generative model for material design based on symmetry inductive bias.
- Implemented the model in PyTorch.
- Achieved the best generation quality and diversity among space-group-conditioned models.
- Led a team of 6.

- **ML for predicting properties of defects in 2D crystals :**

- Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
- Developed code for reproducible parallel running of machine learning experiments.
- Led a team of 3.

Constructor Group

February 2021–present

Machine Learning Consultant

- **AI/ML Consulting :**

- Consulted on machine learning algorithms, deep learning applications, and architectural designs for scientific and industry-led projects.

CERN [Yandex → HSE University]

2014–2022

Researcher

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

- **Generative models uncertainty estimation :**

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

- **Generative models for fast simulation :**

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

- **Machine learning on noisy data :**

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

- **Muon identification at the LHCb experiment at CERN :**

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

- **CatBoost aka fighting biases with dynamic boosting :**

- Set up a distributed system for running experiments for the team with Bash and Python.

- Studied gradient boosting improvements with experiments on toy data.

Education

- Higher School of Economics (HSE University)** 2016–2020
PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).
- Sapienza — Università di Roma** 2016–2020
PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).
- Product Management course at Yandex** Spring 2018
Focused coursework in product strategy and execution.
- Yandex School of Data Analysis** 2013–2015
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.
- Moscow Institute of Physics and Technology**
MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.
BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [20](#)).
- Student council member from 2013 to 2016, leading several dormitory-scale IT projects.
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·
Python (Research coding) ·
PyTorch (Deep learning) ·
C++ (Production) ·
Linux (Administration) ·
Research Writing & Speaking ·
Physics (Deep domain understanding) ·
Leadership & Mentorship

Selected Publications

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). OpenReview, June 2026.
2. **Kazeev, Nikita**, and Phan Bui Nhat Huyen. [Position: Align AI to Our Aspirations, Not Our Flaws](#). Pluralistic Alignment Workshop at ICML 2026, June 2026.

3. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budenny, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). ICML 2026 AI for Science Workshop, May 2026.
4. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
5. Nian Liu, Yuwei Zeng, Ryoji Kubo, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Pengru Huang, Thomas Laurent, Kostya S. Novoselov, and Xavier Bresson. [Composable Crystals: Controllable Materials Discovery via Concept Learning](#). arXiv preprint arXiv:2605.14769, May 2026.
6. Jose Recatala-Gomez, **Nikita Kazeev**, et al. [Generative design of inorganic materials](#). arXiv preprint arXiv:2604.14082, April 2026.
7. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Cl  mentine Fourier, Chaitanya K. Joshi, **Nikita Kazeev**, S  kou-Oumar Kaba, F  lix Therrien, Alex Hern  ndez-Garc  a, Roc  o Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
8. Andrey Okhotin, Maksim Nakhodnov, **Nikita Kazeev**, Mikhail Lazarev, Andrey E. Ustyuzhanin, and Dmitry Vetrov. [MiAD: Mirage Atom Diffusion for De Novo Crystal Generation](#). arXiv preprint arXiv:2511.14426, November 2025.
9. Stephen G. Dale, **Nikita Kazeev**, et al. [AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions](#). arXiv preprint arXiv:2511.20976, November 2025.
10. M. Y. Lukianov, A. Maevskiy, **N. Kazeev**, D. Mylnikov, D. A. Svintsov, K. S. Novoselov, A. Ustyuzhanin, and D. A. Bandurin. [Inverse design of broadband antennas for terahertz devices based on two-dimensional materials](#). *Physical Review Applied* 24, 054079, November 2025.
11. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
12. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). *npj Comput Mater* 9, 113 (2023).
13. Anderlini, L., Chimpoesh, C., **Kazeev, N.**, Shishigina, A., & LHCb collaboration. [Generative models uncertainty estimation](#). *Journal of Physics: Conference Series*, 2023. *I'm the corresponding author.*
14. M. Borisyak and **N. Kazeev**. [Machine Learning on data with sPlot background subtraction](#). *Journal of Instrumentation* 14.08 (2019). *I'm the corresponding author.*
15. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. *I'm the corresponding author.*